

EEAIC - PLL Exercise I

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1 Design of PFD

1.1 Functionality of the PFD (Q2.1)

The core objective of the Phase Frequency Detector (PFD) is to evaluate the timing relationship between two incoming periodic signals—typically the reference clock (*REF*) and the divider feedback (*DIV*)—and generate an error signal. Ideally, the circuit is designed such that its local average output voltage, $\overline{V}_{\text{out}}$, exhibits a linear relationship with respect to the phase difference, $\Delta\Phi$, between its inputs, as highlighted in *Figure 1*.

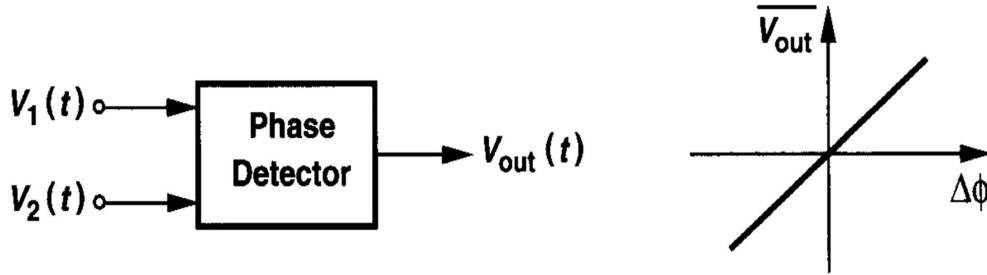


Figure 1: Definition of Phase Detector

Actually, the PFD exhibits dual-mode operational behavior depending on the synchronization state of the loop:

- **Frequency Acquisition Mode (Frequency Detector):**
When the input frequencies are far apart ($\omega_{\text{ref}} \neq \omega_{\text{div}}$), the circuit operates primarily as a Frequency Detector, ensuring a wide acquisition range.
- **Phase Tracking Mode (Phase Detector):**
As the loop brings the frequencies towards equality ($\omega_{\text{ref}} \approx \omega_{\text{div}}$), the PFD seamlessly transitions into a conventional Phase Detector. Once steady-state lock is achieved, the PFD continuously corrects for minor phase perturbations, maintaining a stable, synchronized alignment between the two signals.

The functionality of the PFD is achieved through a straightforward topology, shown in *Figure 2*, that translates edge timing into a voltage correction. The reference and

feedback signals drive the clock inputs of two D-type flip-flops, both of which have their D inputs permanently tied to a high voltage. The Q outputs of these flip-flops act as active-high control signals for the charging and discharging current sources connected to the loop filter capacitance.

When the rising edge of the faster (leading) signal arrives, its corresponding Q output transitions to high, activating its respective current source to either raise or lower the control voltage and adjust the speed of the feedback signal. As soon as the rising edge of the slower (lagging) signal arrives, the second Q output also transitions to high. Because both Q outputs are connected to the inputs of a NAND gate, this simultaneous high state forces the NAND gate output to low, which immediately resets both flip-flops back to $Q = \text{low}$.

This ensures that the current sources are active only during the time window between the two rising edges, making the net charge delivered to the Loop Filter capacitor strictly proportional to the phase difference between the two input signals.

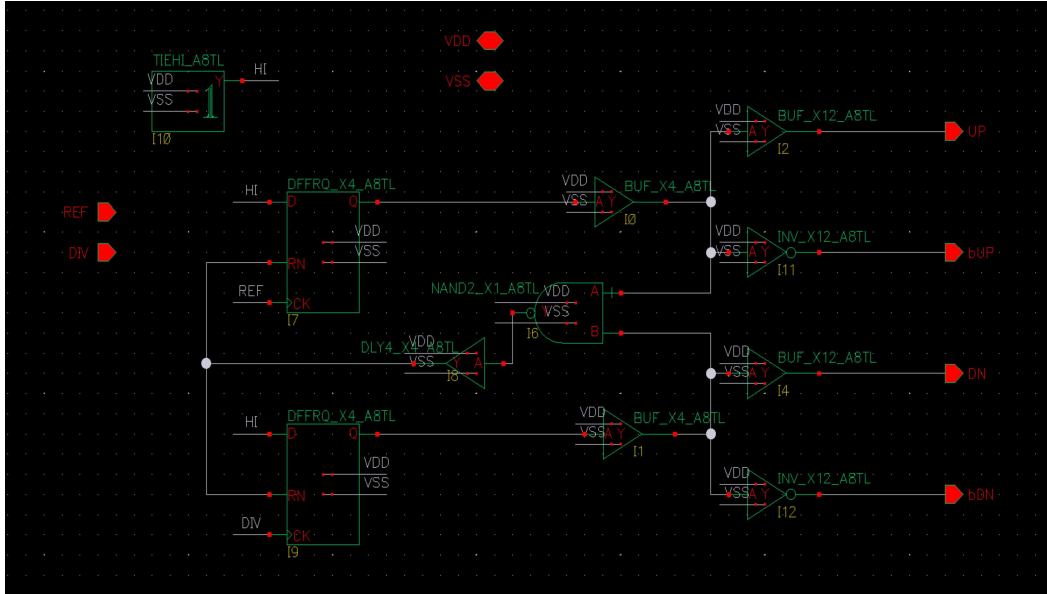


Figure 2: Schematic of the Phase Frequency Detector

1.2 Purpose of Buffers and Delays (Q2.2)

The use of the intermediate BUF_X4 buffers and the dedicated reset delay block DLY_X4, as visible in *Figure 2*, addresses the main physical non-ideality of the PFD: the dead-zone phenomenon.

The deadzone is one of the PFD non idealities that leads to random phase error accumulated, it happens when very small phase differences fail to turn on the charge pump (CP) switches due to finite risetime and falltime.

This deliberate delay ensures that the internal D-flip-flops remain active long enough for the primary control signals to maintain valid logical high levels for the time required by the output current source switches to fully activate ($t_{\text{reset}} > t_{\text{turn-on}}$). The impact of this delay is visible in *Figure 3*, which demonstrates that even after the arrival of the slower edge, the corresponding Q signal remains high for a finite duration that satisfies this criteria.

Simultaneously, the BUF $\times 4$ buffers (where $\times 4$ denotes standard cells with four times the driving capability of a minimum-sized inverter) function as an intermediate scaling stage to optimize the logical effort transition between the relatively weak internal flip-flop outputs and the highly capacitive $\times 12$ final output drivers. This prevents severe degradation of signal slew rates, massive propagation delays, and potential compromises to the timing of the reset loop and the charge pump. Because any asymmetry between these paths would introduce an unwanted static phase offset error, utilizing identical buffers is a critical design requirement that must be verified during testing to ensure both the UP and DN signals experience the exact same propagation delay.

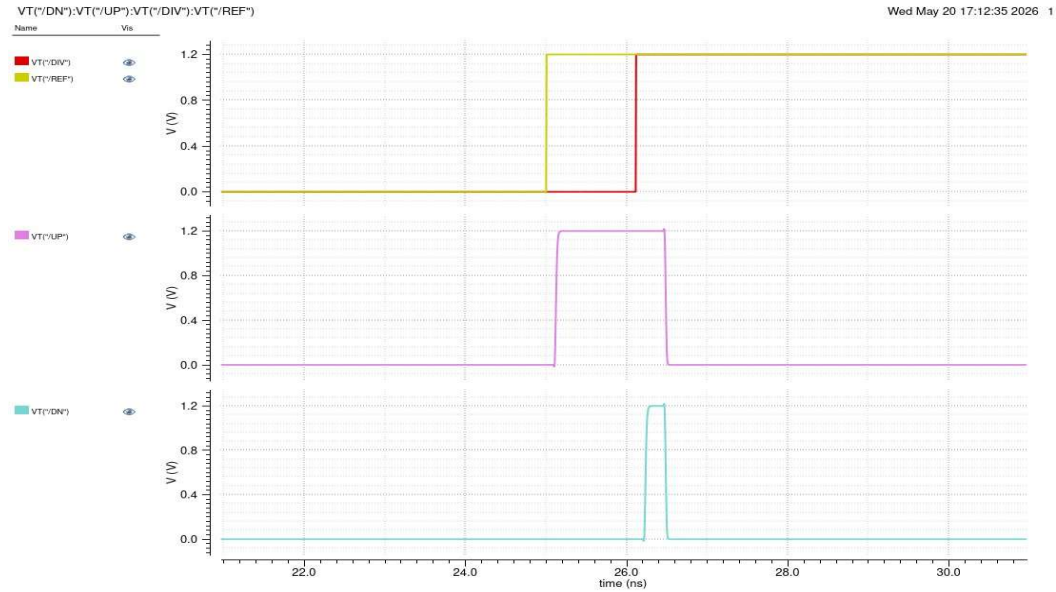


Figure 3: Behavior of the Phase Frequency Detector

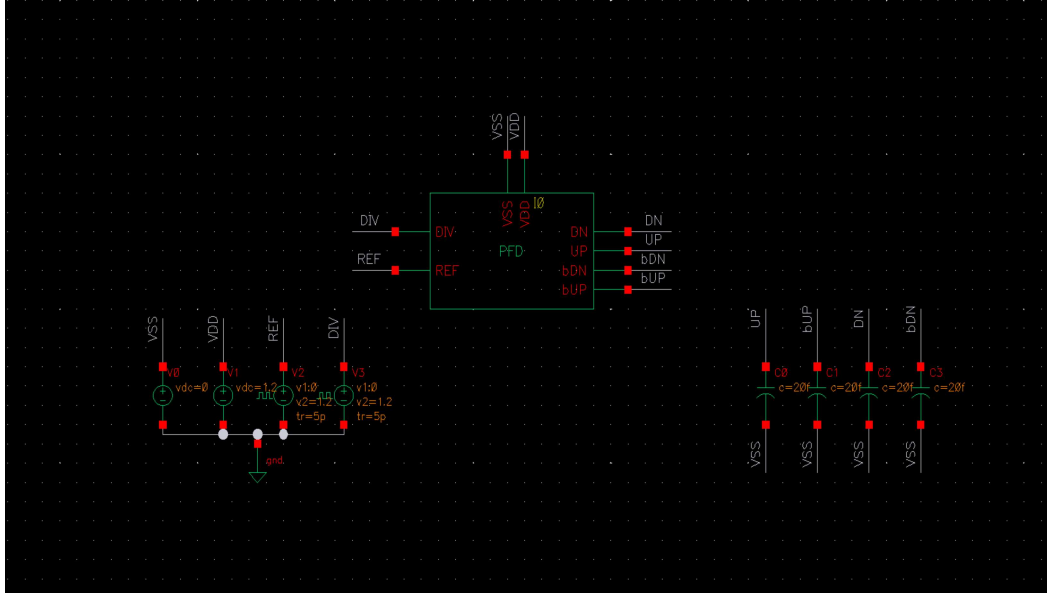


Figure 4: Testbench of the Phase Frequency Detector

1.3 Testing the correct functionality (Q2.3)

Since the non-idealities of the PFD appear during transient events, both verifications require Transient Analysis with specific phase and frequency relationships between *REF* and *DIV*.

- **Testing the DLY block:**

To verify that the delay block successfully eliminates the dead zone, the PFD and charge pump are evaluated using a high-resolution transient phase sweep. The relative arrival time between the reference and feedback clock edges is varied across a narrow window centered around zero phase error. For each step of the sweep, the output current from the charge pump is integrated over a clock cycle to measure the net transferred charge. This allows us to plot the PFD-CP transfer characteristic—net charge versus phase error—and analyze its behavior around the origin. A properly optimized delay block ensures that both current sources have sufficient time to fully turn on and settle, resulting in a strictly linear, monotonic curve with no dead-zone plateaus or slope discontinuities at zero phase error. To guarantee robust operation, this sweep is repeated across all process, voltage, and temperature (PVT) corners, ensuring the reset path remains sufficiently long even under fast corner conditions where the logic gates operate at their maximum speed.

- **Testing the BUF_X4:**

To verify the functionality and performance of these $\times 4$ buffers, a transient simula-

tion is conducted with the circuit operating at its maximum target clock frequency. The required input condition is a fixed, non-zero phase error between the reference and feedback clock signals, which forces the flip-flops to toggle and ensures that the entire buffer chain undergoes a complete logic transition. The testing procedure involves monitoring the voltage waveforms at the input and output nodes of the $\times 4$ stage to measure its propagation delay and evaluate the rise and fall times (slew rates) while driving the full load of the downstream structures. Additionally, the propagation delays through both the reference and feedback buffer paths are compared to confirm near-perfect symmetry, ensuring that no unwanted static phase offset is introduced. Finally, this performance is compared to a benchmark simulation of a buffer-free configuration—where the flip-flops drive the $\times 12$ stages and the NAND gate directly. This comparison isolates and quantifies the improvements in transition sharpness and reset loop speed, validating the eventual necessity of the intermediate buffer stage.

2 Design of Charge Pump

2.1 Dump Branch (Q3.1)

In a simplified charge pump design, the UP and DN signals dynamically turn on and off the current sources by switching the transistors directly. However, this architecture suffers from major non-idealities: charge injection and clock feedthrough.

Indeed, the charge injection is due to the fact that in the inversion layer of each MOSFET there is a certain amount of charge accumulated; when the transistor goes off, these charges are approximately equally split between Source and Drain. The clock feedthrough is due to the presence of parasitic capacitances that directly couples the input to the output V_{CP} . Both of these non-idealities lead to severe spurs in the PLL output.

There are many possible remedies for minimizing these issues, e.g. introducing dummy switches or even a more popular architecture to add a dump branch.

The dump branch solution is a popular current-steering architecture that consists in keeping the current sources continuously active and simply steering the current away from the loop filter when it is not needed. Because the current sources never turn off, the parasitic capacitances at the internal nodes remain fully charged to their nominal operating voltages. Consequently, the current response becomes faster and sharper, and the total turn-on time of the charge pump drops dramatically.

A smaller required pulse width means less redundant power consumption, less injected noise during the overlap period, and thus an overall reduced dead zone. However, we must pay attention to the fact that now the power consumption is no longer purely dynamic but has become static. Even if this is a significant drawback, the trade-off is almost always worth it: the reduction in reference spurs and the effective shrinking of the dead zone justify the increased power budget. Crucially, this architecture only resolves the issue when coupled with a voltage-matching unity-gain amplifier. By actively forcing

the dump node voltage to track V_{ctrl} , the amplifier prevents charge-sharing transients when the current is redirected, thereby eliminating the underlying mechanism of the switching spurs.

When the switches flip, the parasitic charges on the gates are mostly absorbed by the complementary switching action rather than dumped onto the loop filter, dramatically reducing the size of the voltage glitches on V_{ctrl} .

In a conventional charge pump, when a switch is turned off, the drain voltage of the current source transistor drifts toward the supply rail, pulling the transistor out of the saturation region. When the switch turns back on, the transistor has to transition from a non-saturated state back into saturation, causing a massive current spike. With a dump branch, the current source transistor always stays in saturation. It continuously pumps the exact same current (I_{CP}); the circuit just changes where that current goes. Because the transistors never have to recover from being squeezed out of saturation, the charge pump responds instantly and cleanly to the PFD pulses.

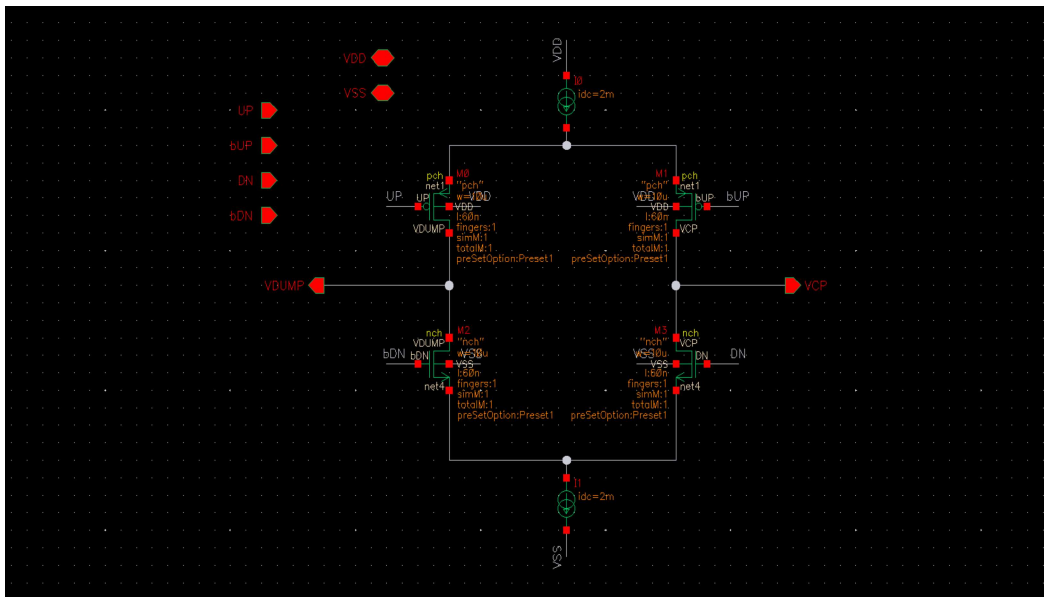


Figure 5: Simplified Schematic of the Charge Pump

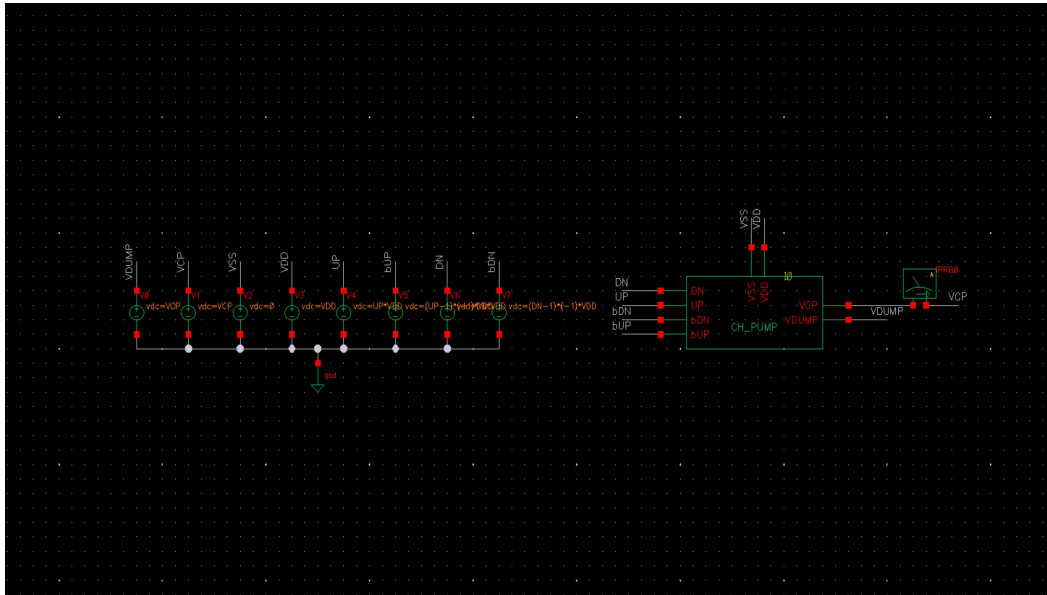


Figure 6: Testbench of the Charge Pump

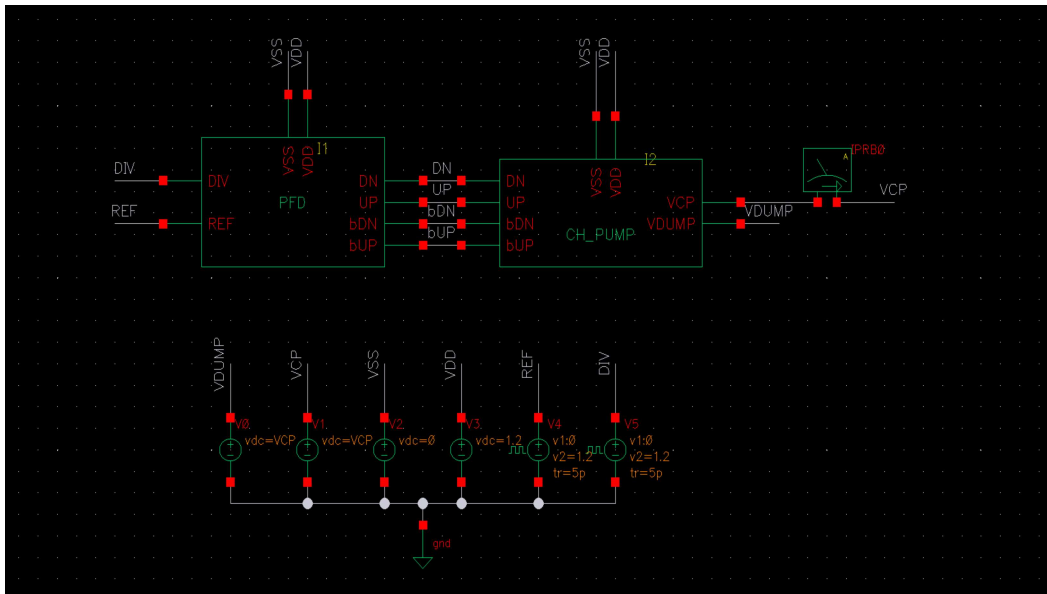


Figure 7: Schematic of the Phase Frequency Detector and Charge Pump

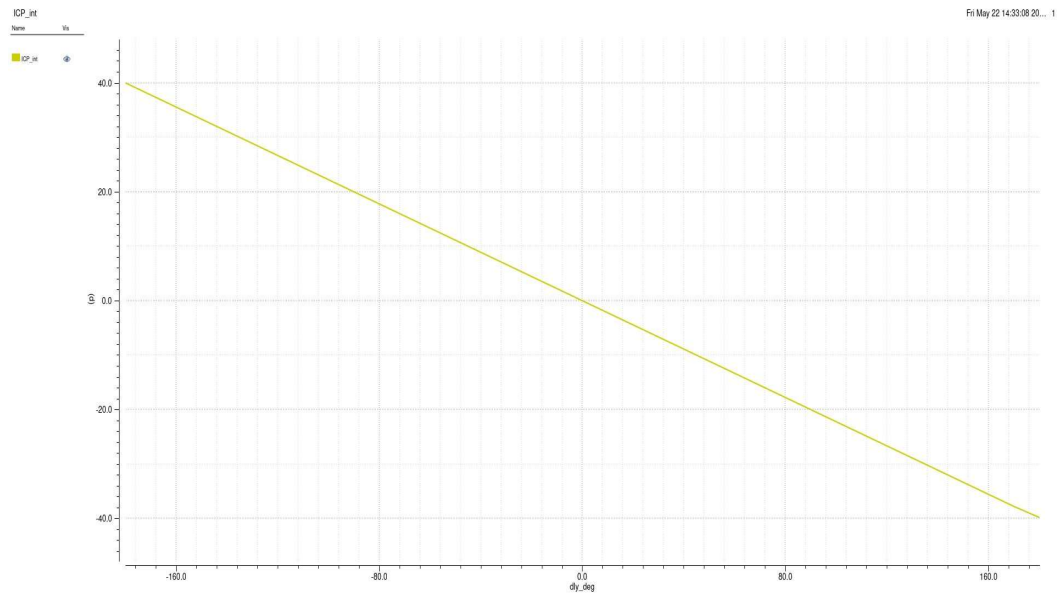


Figure 8: Output of ideal CP

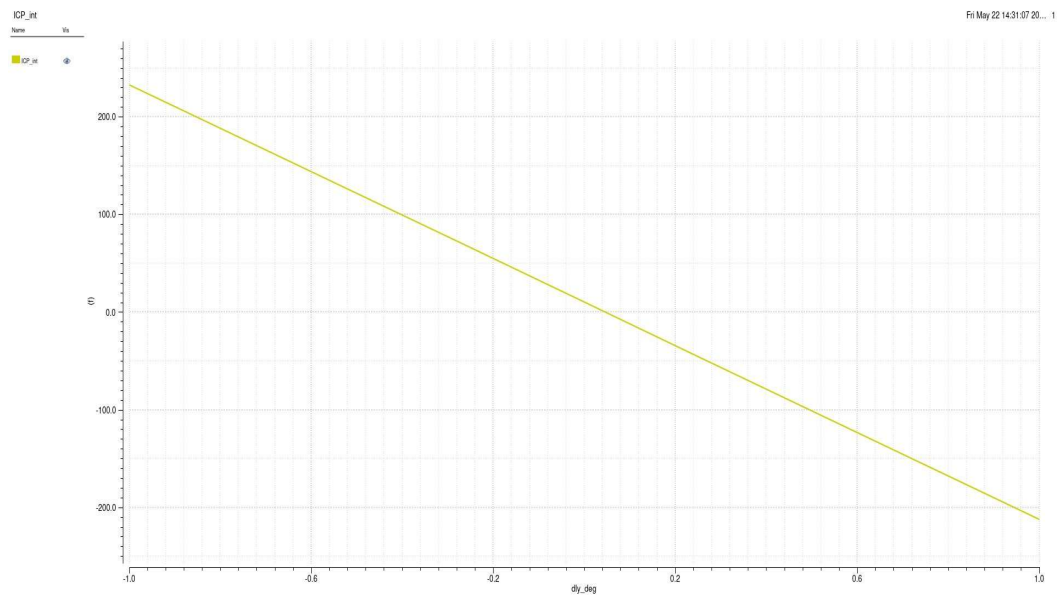


Figure 9: Deadzone of ideal CP

3 Design of real Charge Pump

3.1 Charge Pump Design and Optimization Flow (A4.1)

Architectural Evolution and Bias Topology

The optimization process of the charge pump followed an iterative approach, primary focused on simultaneously satisfying static current constraints and spectral noise requirements. The initial architecture, based on a conventional current mirror topology, was found to be inadequate. Meeting the target specifications would have demanded excessive transistor sizing, severely penalizing silicon area and increasing parasitic capacitances.

To overcome these limitations, a more sophisticated structure was implemented, inspired by literature *Design of Analog CMOS Integrated Circuits* of Razavi, as illustrated in *Figure 10*. Through this configuration, an optimal aspect ratio (W/L) was determined, successfully maintaining a bias current of $20\mu\text{A}$ in the reference branch and an output current of 2mA in the final branch.

To minimize output noise, current mirrors were implemented using thick-gate devices, which exhibit significantly lower gate leakage. In contrast, the remaining transistors that represent the switches utilized thin-gate geometries to optimize performance.

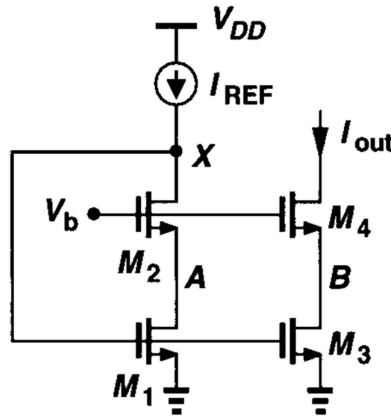


Figure 10: Modification of cascoded mirror for low-voltage operation

To minimize the overall noise contribution, the current of the intermediate branch (I_{int} , depicted in *Figure 18*) was maximized to a nominal value of $274\mu\text{A}$. This design choice ensures that, even under worst-case operating conditions, the total current consumption remains bounded at 2.296mA , fully complying with the maximum power budget specification of 2.3mA .

Characterization Testbenches

Throughout the entire design process flow, dedicated testbenches were continuously utilized to characterize individual transistors as well as differential/mirror transistor pairs. These testbenches allowed for the precise simulation of expected physical behaviors, ensuring that the target overdrive voltages (V_{ov}) and drain-source voltages (V_{ds}) were accurately achieved under nominal biasing conditions.

This verification methodology was applied iteratively across all design phases to validate transistor-level operating regions prior to full-circuit integration. The corresponding schematic configurations for these characterization testbenches are illustrated in *Figure 11* and *Figure 12*.

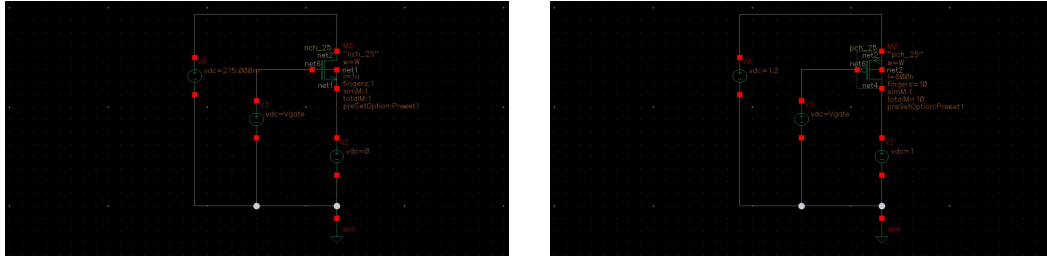


Figure 11: Testbench for the single nMOS and pMOS

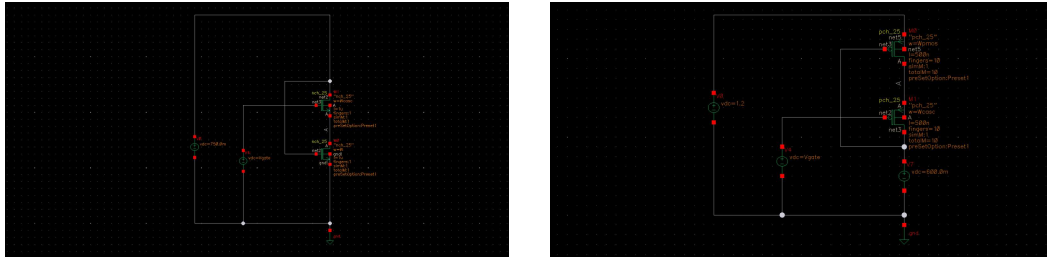


Figure 12: Testbench for the cascaded nMOS and pMOS

High-Frequency Noise Mitigation via Pseudo-Resistor

Although the aforementioned configuration met the static current constraints, the power spectral density (PSD) of the output current noise remained unacceptably high. To suppress this contribution, a pseudo-resistor was integrated between the gate nodes of the current mirrors.

The implemented pseudo-resistor structure (shown in *Figure 13*) utilizes the identical topology previously discussed for the Fully Differential Amplifier. Among the analyzed

trade-offs, "Configuration 1" was selected because it exhibits a more linear behavior that closely emulates an ideal passive resistor.

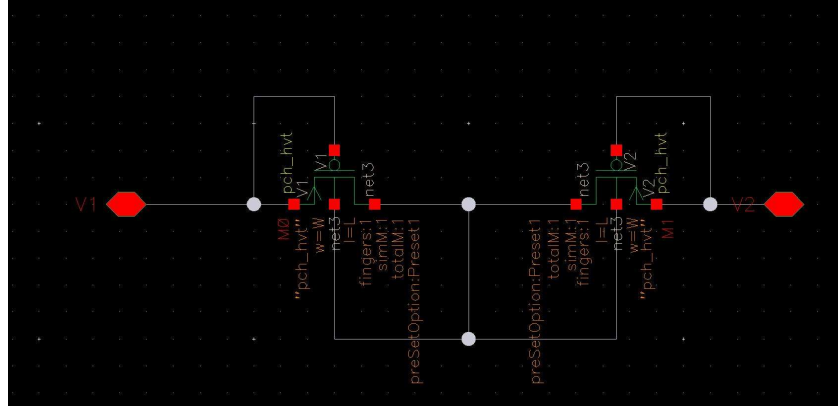


Figure 13: Configuration 1 of the Pseudoresistor

The interaction between the pseudo-resistor and the parasitic gate capacitances (C_{gg}) of the respective transistors implements a low-pass filter (LPF). This filter introduces an averaging effect on the noise fluctuations, significantly attenuating high-frequency spectral components and leading to a substantial improvement in the total integrated root-mean-square noise.

Flicker Noise ($1/f$) Optimization

From the current noise PSD plot, it was observed that the flicker noise corner frequency (f_c) was initially located too far into the high-frequency spectrum, at approximately 450 kHz. Given the substantial margin available (roughly $350 \mu\text{A}^2/\text{Hz}$) with respect to the thermal noise floor specification of $1 \text{ zA}^2/\text{Hz}$, design efforts were directed toward shifting f_c to lower frequencies.

By equating the thermal noise PSD to the flicker noise PSD, the corner frequency can be analytically expressed as:

$$f_c = \frac{K \cdot g_m}{C_{ox} \cdot W \cdot L \cdot 4kT \cdot \gamma}$$

The analytical expression highlights that the channel area ($W \cdot L$) is inversely proportional to f_c . To lower the corner frequency without altering the bias conditions, current density, or mirror ratios, both the channel widths (W) and lengths (L) of the transistors were scaled up proportionally, thereby preserving the original aspect ratio (W/L). This geometric multiplication is explicitly highlighted in the schematic of *Figure 18* by the scaling factor "4".

Simulation Results

Following the final sizing implementation, the definitive circuit architecture and sizings can be extracted from *Figure 14* and *Figure 18*. Simulation results confirm that all project requirements are met:

- **Static Currents:** The currents measured in all branches adhere strictly to the targeted design values and consumption limits, as highlighted in *Figure 15* and also in *Figure 18*.
- **High-Frequency Noise:** At the reference frequency of 2 MHz, the current noise spectral density is approximately $528 \text{ yA}^2/\text{Hz}$ (*Figure 16*), which is safely below the maximum allowable specification of $1 \text{ zA}^2/\text{Hz}$ ($1000 \text{ yA}^2/\text{Hz}$).
- **Flicker Noise Corner:** To extract the flicker noise corner frequency, the thermal noise floor was first established at the sufficiently high frequency of 2 MHz where white noise is completely dominant and flicker noise is negligible. By shifting this baseline thermal floor upward by +3 dB, corresponding to a doubling of the dominant background noise power, the intersection with the total noise curve defines the effective flicker corner, as highlighted in *Figure 17*. The extracted corner frequency is located at 33.6 kHz, which fully satisfies the design specification mandating a corner frequency below 50 kHz.

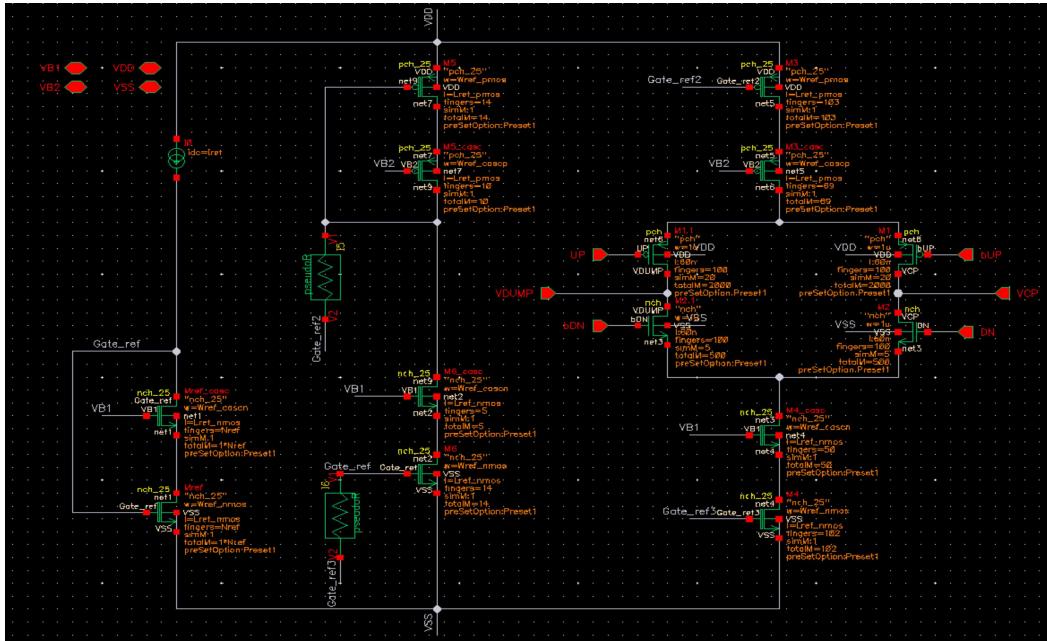


Figure 14: Schematic of a real CP

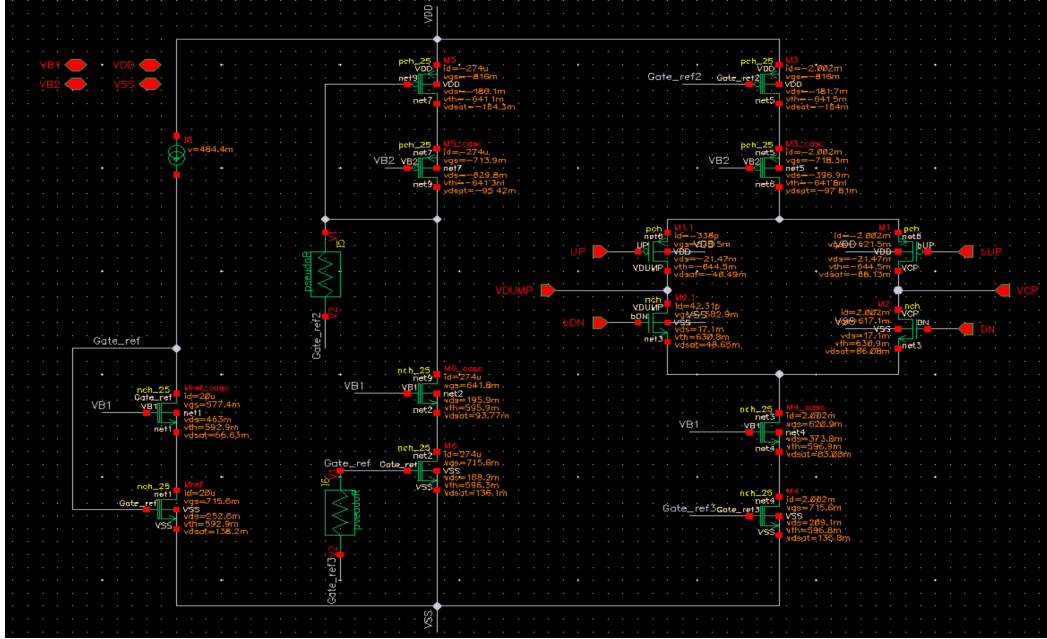
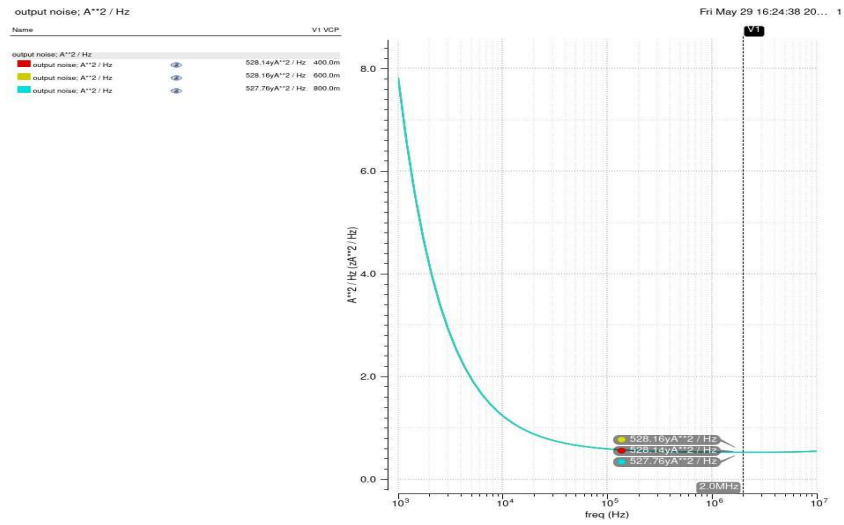
Figure 15: DC operating points of the real CP in the case of $V_{CP} = 0.6V$ 

Figure 16: Output Noise for Noise Thermal Floor

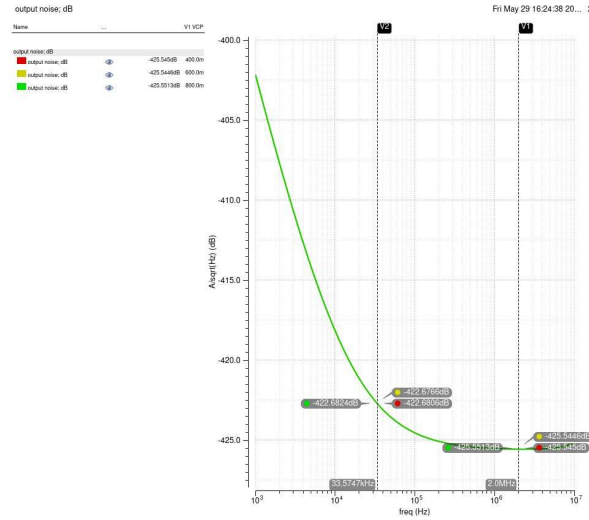


Figure 17: Output Noise in dB scale for Flicker corner

PHDETECTOR_zTEST_CP_REAL_1	
Simulator	spectre
Analyses	
☒ dc	t
☒ noise	1k 10M Automatic Start-Stop
Click to add analysis	
Design Variables	
VDD	1.2
UP	1
DN	1
Iref	20u
Nref	1
Lref_nmos	1u*4
Lref_pmos	0.5u*4
Wref_cascp	60u*4
Wref_pmos	12u*4
Wref_cascn	60u*4
Wref_nmos	8.5u*4
VB1	830m
VB2	300m
VCP	0.4:0.2:0.8
W	5u
L	1u
Click to add variable	
Parameters	

Point	Test	Output	Nominal
Filter	Filter	Filter	Filter
Parameters: VCP=400m			
1	PHDETECTOR_z...	output noise; A*...	
1	PHDETECTOR_z...	I_DN	2.001m
1	PHDETECTOR_z...	I_UP	2.001m
1	PHDETECTOR_z...	I_int	274u
1	PHDETECTOR_z...	output noise; dB	
Parameters: VCP=600m			
2	PHDETECTOR_z...	output noise; A*...	
2	PHDETECTOR_z...	I_DN	2.002m
2	PHDETECTOR_z...	I_UP	2.001m
2	PHDETECTOR_z...	I_int	274u
2	PHDETECTOR_z...	output noise; dB	
Parameters: VCP=800m			
3	PHDETECTOR_z...	output noise; A*...	
3	PHDETECTOR_z...	I_DN	2.002m
3	PHDETECTOR_z...	I_UP	2m
3	PHDETECTOR_z...	I_int	274u
3	PHDETECTOR_z...	output noise; dB	

Figure 18: Up and Down currents and design parameters

3.2 Noise Contributors and Transistor Sizing Strategy (A4.2)

As visible from the noise summary report illustrated in *Figure 19*, the dominant contributions to the total output noise originate from the transistors carrying the highest currents. To ensure accurate current copy and matching, the current mirror transistors

are designed to operate strictly in the strong inversion (SI) regime. In this operating region, the resulting current noise PSD can be analytically modeled as:

$$S_i(f) = \left(\frac{K}{C_{ox}WL} \cdot \frac{1}{f} + 4kT\gamma \frac{1}{g_m} \right) \cdot g_m^2 = \frac{K \cdot g_m^2}{C_{ox}WL \cdot f} + 4kT\gamma g_m$$

In the strong inversion regime, the transconductance is related to the drain current and the overdrive voltage by: $g_m = \frac{2I_{DS}}{V_{ov}}$

Device	Param	Noise Contribution	% Of Total
/I2/M4	id	2.72446e-15	53.63
/I2/M3	id	2.09403e-15	41.22
/I2/M4_casc	id	1.69021e-16	3.33
/I2/M3_casc	id	6.6972e-17	1.32
Integrated Noise Summary (in A^2) Sorted By Noise Contributors			
Total Summarized Noise = 5.0803e-15			
No input referred noise available			
Total noise by type:			
Flicker: 0.00000E+00			
Thermal: 5.05449E-15			
Shot: 0.00000E+00			
The above noise summary info is for noise data			

Figure 19: Charge Pump noise summary evaluated at $V_{CP} = 0.6$ V.

Since the output branch current is fixed at 2 mA by the project specifications, minimizing the transconductance g_m , and consequently minimizing both thermal and flicker noise contributions, requires maximizing the transistor overdrive voltage V_{ov} .

To eliminate systematic mismatches across the parallel branches of the topology, a rigorous unit-transistor sizing strategy was adopted. The primary mirror devices (M_{ref} , M_6 , M_4 for the NMOS side, and M_5 , M_3 for the PMOS side) share identical unit dimensions (W_{unit}/L_{unit}), ensuring that all current ratios are established exclusively by scaling the number of fingers and the multiplicity factor (M). The exact same approach was applied to the cascoded devices. By enforcing this layout-aware design methodology, every branch is constructed strictly as an aggregate replica of the same reference unit cell. Modifying only the fingers and multiplier values allows the current mirroring factor to be scaled precisely without altering the intrinsic layout profile or the core electrical behavior of the devices. This preserves identical local matching properties and unit-cell area across the entire circuit matrix, maximizing immunity to process variations.

The primary trade-off of maximizing the overdrive voltage is the reduction of the available voltage swing at the charge pump output. Increasing V_{ov} inherently raises the minimum required saturation voltage ($V_{DS,sat}$), thereby compressing the usable dynamic range of the control voltage V_{CP} . For this standalone design stage, V_{CP} is fixed via an ideal voltage source, allowing for an optimal maximization of V_{ov} to optimize noise performance. However, during full-system integration, this careful tradeoff of V_{ov} between the charge pump and the VCO has to be considered.

Finally, the sizing of the switching transistors (M_1 , $M_{1.1}$, M_2 , and $M_{2.1}$) was optimized by establishing a strategic compromise between a small on-resistance ($R_{ds,on}$) and minimized parasitic capacitances. While these switching devices do not contribute significantly to the total output thermal noise power, their geometric dimensions heavily dictate both the static headroom and the dynamic accuracy of the charge pump. On one hand, keeping $R_{ds,on}$ sufficiently low is mandatory to prevent excessive voltage drops across the switches, thereby safeguarding the vertical headroom and ensuring that the cascode transistors remain firmly saturated. On the other hand, keeping the device sizes bounded minimizes the parasitic gate capacitances, which effectively suppresses charge injection and clock feedthrough artifacts during fast switching transitions. Therefore, rather than simply maximizing the W/L ratios, the switches were sized to achieve an optimal balance that minimizes conduction losses without introducing detrimental capacitive penalties.

3.3 V_{ov} Choice (A4.3)

To analytically derive the minimum required overdrive voltage, the noise analysis can be tightly focused on the output branch of the charge pump topology. Since this branch conducts the most significant fraction of the total current compared to the reference and intermediate branches, its contribution to the overall thermal noise is heavily dominant in a first-order approximation, as highlighted in *Figure 19*.

Considering the noise contributions from the switching transistors and cascoded devices negligible, as clear in *Figure 19*, the total thermal current noise PSD at the output node is dictated exclusively by the active transistors in the final stage. In a theoretical scenario where the nMOS/pMOS main mirrors (M_3 and M_4), contribute simultaneously and share an identical overdrive voltage, the total thermal noise PSD would be modeled as:

$$S_{I,\text{thermal}} \approx 2 \times (4k_B T \gamma g_m) = 8k_B T \gamma g_m \leq S_{\text{target}} = 1 \text{ zA}^2/\text{Hz}$$

$$V_{ov} \geq \frac{16k_B T \gamma I_D}{S_{\text{target}}}$$

By substituting the numerical parameters into the corrected equation and considering a standard long-channel thermal noise coefficient $\gamma = 2/3$, the calculation reveals that a minimum overdrive voltage of approximately 88 mV is required to lower the global transconductance g_m sufficiently to satisfy the project specifications.

In our final design, this target constraint V_{ov} is successfully satisfied. However, it must be noted that the individual noise contributions of the transistors of the last branch vary due to asymmetric sizing and distinct operating parameters. Consequently, this simplified analytical derivation served primarily as a first-order estimation to establish an initial baseline for the expected design metrics.

3.4 Equalizing Node Voltages (A4.4)

A voltage mismatch between V_{CP} and V_{DUMP} nodes introduces significant non-idealities during dynamic current-steering operations. First, it causes a **static current mismatch** due to channel length modulation; despite the high output impedance of the cascode structures, the finite R_{ds} of the transistors leads to varying V_{DS} levels depending on whether the current is steered to the output or the dummy branch. Second, it induces **dynamic charge sharing**. The internal parasitic capacitances at the intermediate nodes between the cascodes and the switches must rapidly charge or discharge to accommodate the shifting potentials whenever a branch switches between V_{DUMP} and V_{CP} , translating into a transient current glitch injected directly into the output.

These non-idealities heavily degrade the performance of the overall PLL synthesizer in two primary ways:

- **Reference Spurs:** When the PLL achieves lock, the PFD periodically activates the charge pump for brief intervals to prevent the dead-zone. If $V_{DUMP} \neq V_{CP}$, each periodic turn-on event forces a deterministic current glitch onto V_{CP} . This periodic disturbance creates severe reference spurs in the frequency spectrum.
- **Static Phase Error:** The asymmetric current spikes and underlying current unbalance alter the net average charge delivered to the loop filter. To compensate for this error and maintain a net zero average current during steady-state lock, the PLL settles with a permanent time offset between the reference and feedback clocks, introducing an unwanted static phase error at the PFD inputs.

To dynamically mitigate this mismatch, an **unity-gain buffer** can be integrated between V_{CP} and V_{DUMP} nodes, as shown in *Figure 20*. Due to the virtual short-circuit property of the high-gain amplifier, V_{DUMP} is actively regulated to track V_{CP} across the entire output voltage swing, neutralizing both channel length modulation and charge-sharing glitches. Implementing this buffer introduces specific design trade-offs, namely increased static power consumption from the amplifier bias network, stability constraints under capacitive switching loads, and additional input parasitic capacitance loading the sensitive V_{CP} node, which must be carefully minimized to avoid affecting the loop filter dynamics.

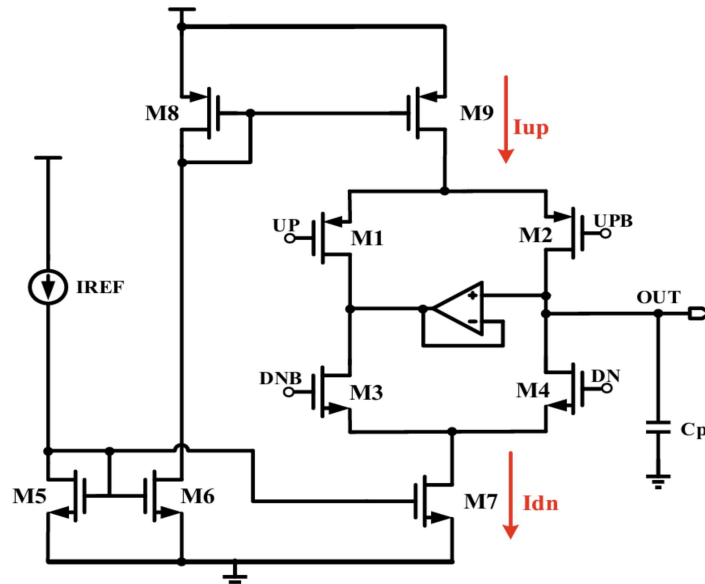


Figure 20: Schematic of a conventional Charge Pump

Energy-efficient Analog Integrated Circuits

PLL – Exercise 1

Giorgio Cristiano

Prof. Taekwang Jang

1. Introduction

In this exercise session we are going to explore the design procedure for a charge pump PLL (CPPLL), depicted in Figure 1. We will start from the design of the in-band contributors, namely phase detector and charge pump. Then, we will move to the VCO, and finally we will put everything together on MATLAB to better analyze the closed loop behaviour. The PLL will multiply a reference frequency of 25MHz by 80, obtaining an output frequency of 2GHz. Here we will focus on phase detector and charge pump.

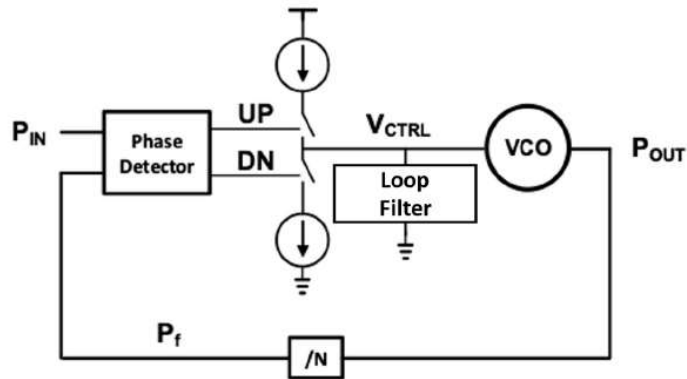
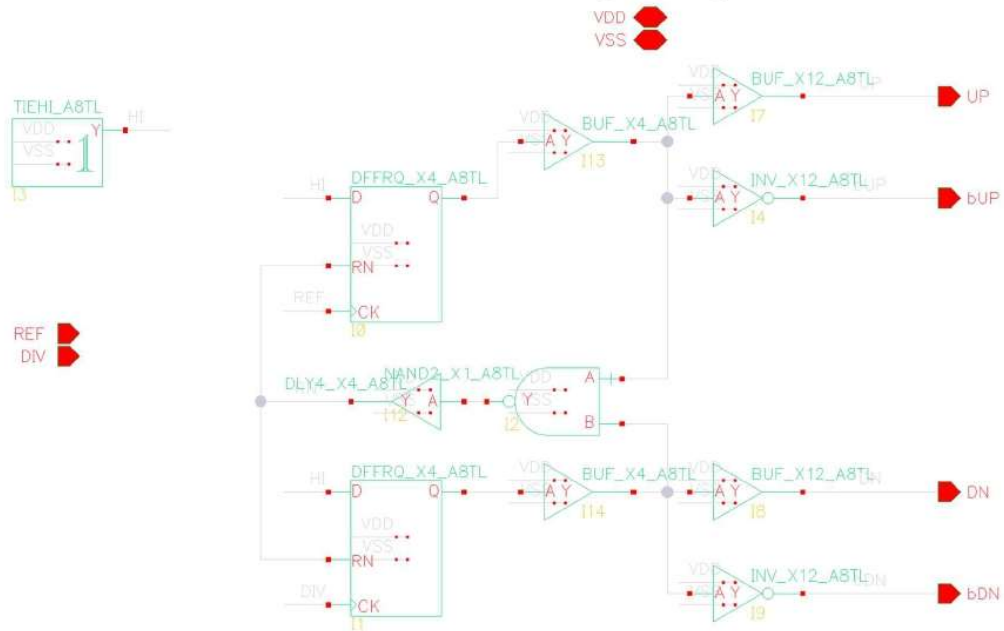


Figure 1: Block diagram of CPPLL

2. Design of PFD

The phase detector is a key component, capable of detecting the phase difference between two signals. In a PLL, this is necessary to translate the phase difference between reference and fed

In this design we will use a PFD based on two FFs. In a new library called PHDETECTOR, create a new cellview, call it PFD, and draw the schematic depicted in Fig.2.



Let's now test this PFD. Create a symbol for the PFD cellview. Then, create a new cellview called `zTEST_PFD`, and in the schematic instantiate the PFD just created. Add a 20fF loading capacitor at each at the outputs; this is used to mimic the loading of the block, as in normal operation it will be connected to the charge pump. Fix VSS at 0V, and VDD at 1.2V. For REF and DIV, we want to create a testbench in which the phase delay for the two signals can be varied.

Q2.1: Please describe the functionality of the PFD.

Q2.2: What is the purpose of the Buffers and Delays (I12, I13, I14 in the picture above)? Why are those components needed?

Q2.3: Please elaborate on how to test the correct functionality of the instances mentioned in the question Q2.2. Do you need a specific input condition to test them?

3. Design of Charge Pump

Create a new cellview, and call it CH_PUMP, and draw the schematic depicted in Fig. 5. In this schematic, the charge pump is simplified by using some real current sources as source and sink currents, while switches connected to the UP, DN signals coming from the PFD dictate when current will be fed/withdrawn from the node VCP. Create a symbol.

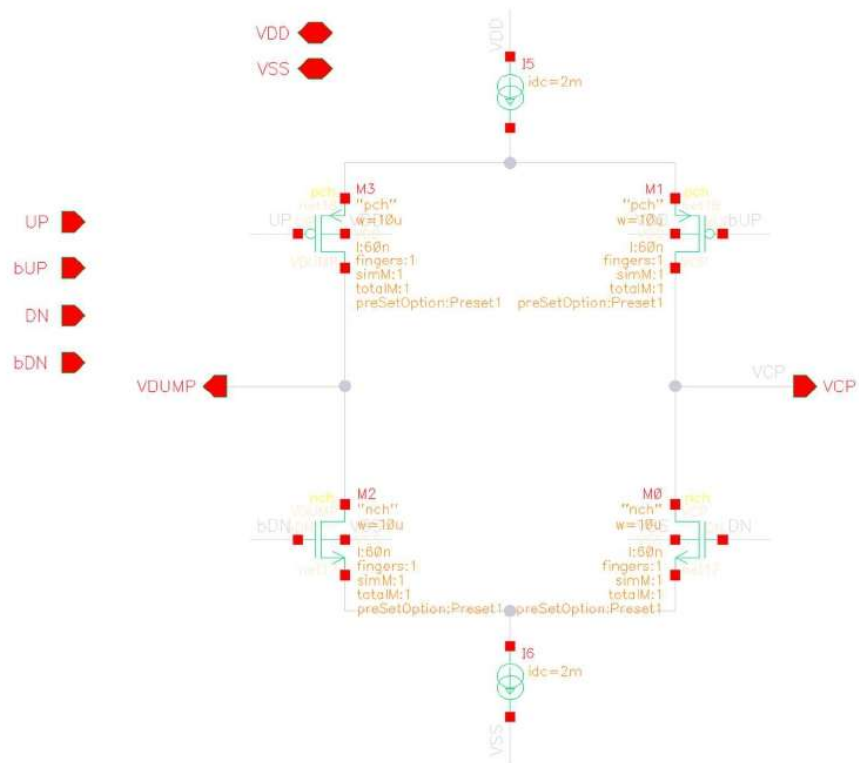


Fig. 5: Simplified schematic of charge pump

Q3.1: Why am I adding the dump branch connected to VDUMP? What do you think would be the drawback of not having it? (Consider the case with a real mirror, instead of the ideal current source used above).

4. Design of real CP

Now you will apply what you have learnt so far to design a charge pump without ideal blocks. The three biggest issues that will arise the moment in which you will replace the ideal sources with mirrors are source and tail matching, CP noise, and flicker noise.

A4.1: Design a CP with 2mA of nominal current generated from a single ideal current source of 20uA, that meets the following specs:

1. Max UP-DN current mismatch of 0.5% (10uA)
2. Mismatch constraint respected at different VCP voltages (0.4V, 0.6V, 0.8V)
3. Max current consumption of the whole block 2.3mA
4. Noise thermal floor below $1e-21$ V²/Hz
5. Flicker corner before 50kHz

Hint: a simple current mirror will probably not be enough. Do you know techniques for increasing the impedance of the output branch from the AIC course?

Hint2: The diode connected side of the mirrors would also contribute to the output noise, however we can filter them out with a simple RC. Making a large C is expensive, as it would require a large amount of area, and it is thus impractical. On the other hand, a large R can be implemented with a much smaller area investment using pseudo-resistors, or duty cycled resistors. However, a large R would lead to a large voltage drop due to gate leakage. This will create a large mismatch between source and tail currents, and should thus be avoided. The easiest solution to this is to simply design the mirrors with thick gate transistors (nch_25 and pch_25), which have much smaller gate leakage. Note that you should have just enough headroom with $V_{DD}=1.2V$, so you do not need to increase it.

A4.2: What are the biggest noise contributors? How would you minimize their contribution? What are the tradeoffs? Explain qualitatively your design choices for the key transistors of the block.

A4.3: What is the V_{ov} needed to achieve the target noise floor? (Consider $\gamma=2/3$)

A4.4: Up to now we are connecting VCP and VDUMP to the same potential artificially in the TB. This, however, cannot be done in reality, as we do not know the exact voltage at which the PLL will lock beforehand. What do you think will happen if VDUMP and VCP are not matched? How will it reflect on the PLL? What could you add to the circuit to make sure that the two nodes are matched? Explain qualitatively.